Brain Tumor Detection using CNN

Stream: Machine Learning

A Comprehensive Approach to
Automated Brain Tumor Detection



Introduction

Overview of the project:

- This project aims to develop and deploy a Convolutional Neural Network (CNN) for detecting brain tumors from MRI images.

Importance in the current context:

- Early and accurate detection of brain tumors is crucial for effective treatment and improved patient outcomes.

Objectives of the CCP:

- To create an automated system for brain tumor detection with high accuracy and reliability.

Problem Statement



Brain tumors are life-threatening, and their detection is often challenging due to the complexity of MRI images.



Manual analysis of MRI scans by radiologists is time-consuming and subject to human error.



An automated detection system can assist radiologists, reduce diagnosis time, and improve accuracy.

Proposed Solution

Utilize Convolutional Neural Networks (CNNs) to automatically identify and classify brain tumors from MRI images.

Key components of the solution:

1. Preprocessing of MRI images.

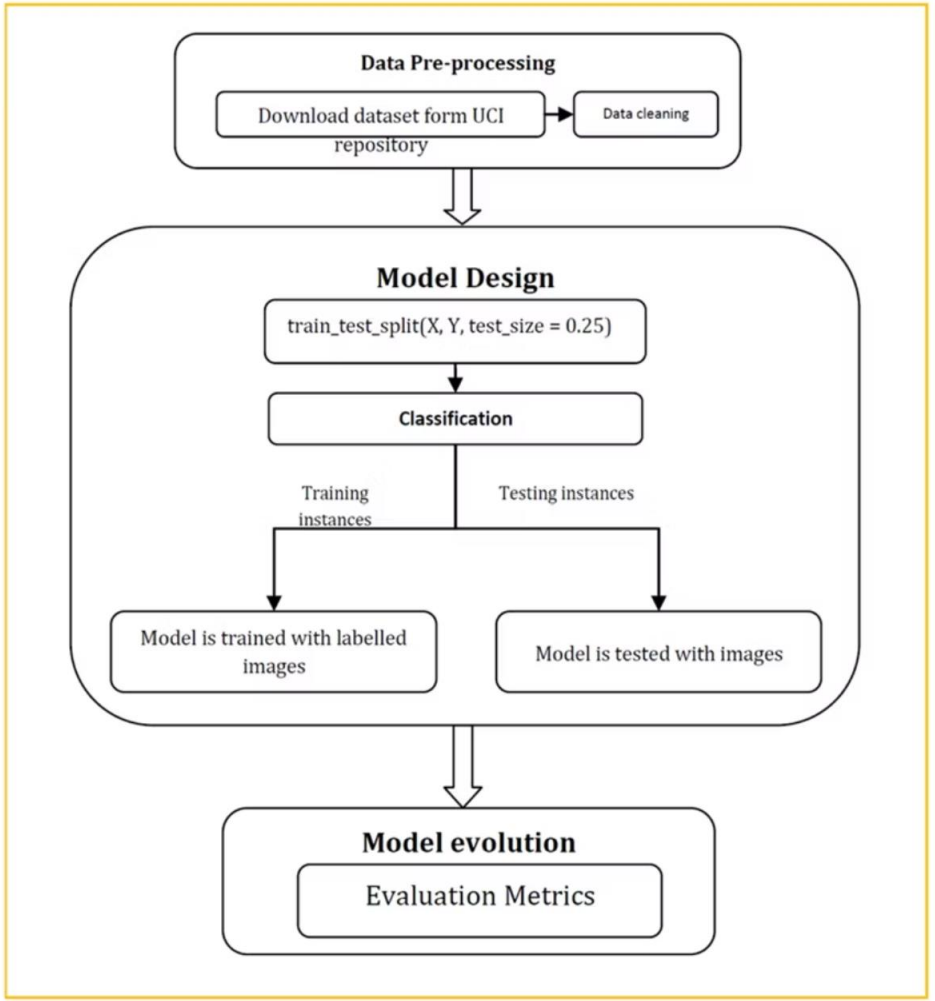
2. CNN architecture design.

3. Model training and evaluation.

4. Model Deployment.

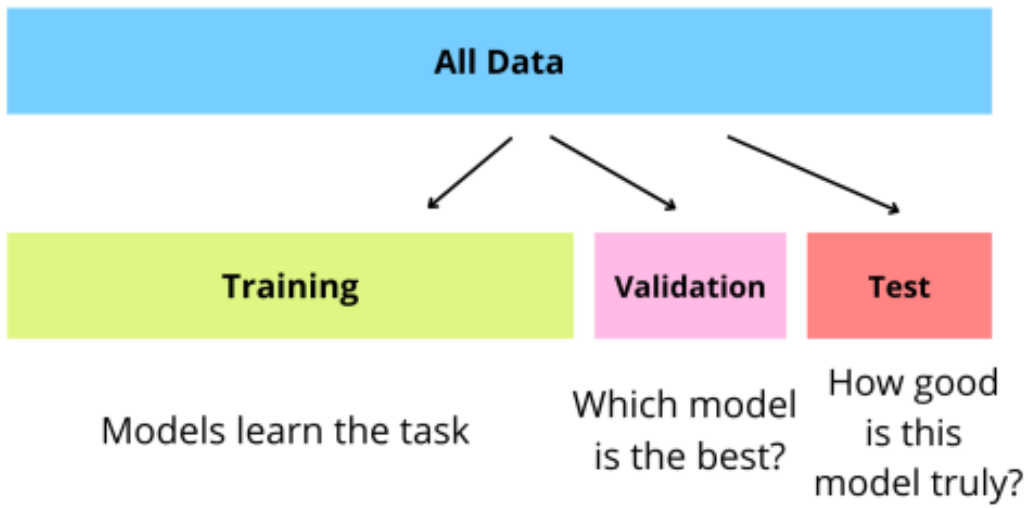
Implementation of source code

- Data Pre-Processing
- Model Training
- Model Evaluation
- Web App

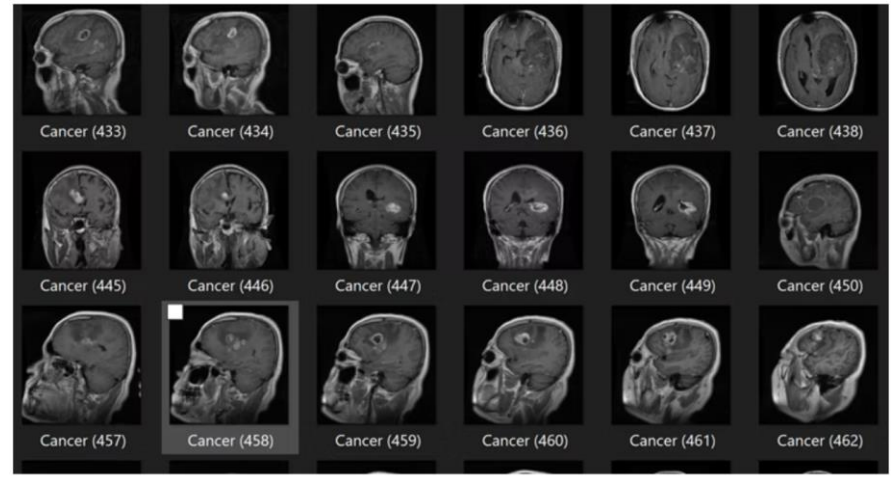


I. Data Pre-Processing

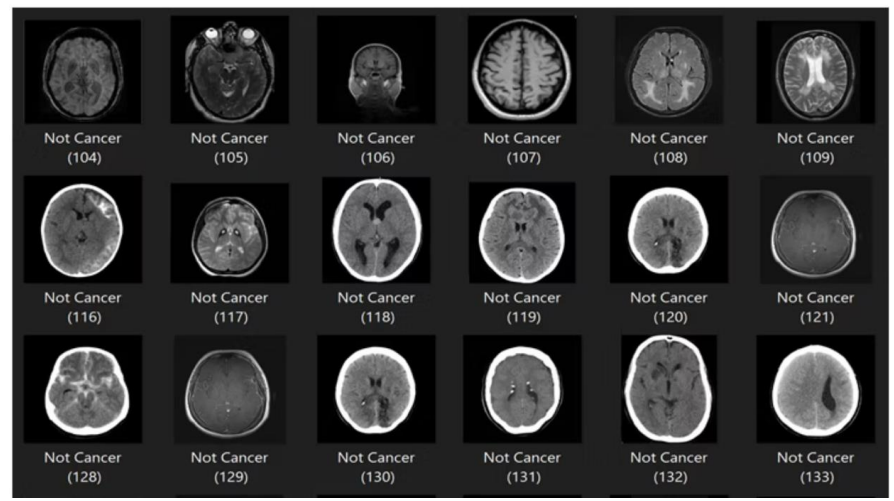
- Collect the images from UCI and Kaggle data repositories
- Split the images in to 3 segments
- Training data – 70%
- Testing data – 15%
- Validation data – 15%



Brain Tumor Images



Non-Brain Tumor Images



II. Model Training

- Importing Libraries

```
# importing Deep learning Libraries
import tensorflow as tf # for building neural networks
from tensorflow import keras # for building neural networks
from tensorflow.keras.models import Sequential # for building sequential neural networks
from tensorflow.keras.optimizers import Adam, Adamax # for optimizing neural network training
from tensorflow.keras.preprocessing.image import ImageDataGenerator # for generating images for training
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Activation, Dropout, BatchNormalization
from tensorflow.keras import regularizers # for regularization techniques
```

- Train the model with epoch-30 , steps for each epoch – 10.
- Set the early stopping and Model Check Points.

```
epochs = 30   # number of all epochs or passes in training

history = model.fit(x= train_gen, epochs= epochs, verbose= 1, validation_data= test_gen,
                    validation_steps= None, shuffle= False)
```

```
Epoch 1/30
26/26 [==============================] - 24s 275ms/step - loss: 8.2593 - accuracy: 0.7525 - val_loss: 9.1887 - val_accuracy: 0.6078
Epoch 2/30
26/26 [==============================] - 4s 169ms/step - loss: 7.4990 - accuracy: 0.7970 - val_loss: 9.2401 - val_accuracy: 0.6863
```

II. Model Training

- After training save bestmodel.h5 file

```
Epoch 8/30
10/10 [==============================] - ETA: 0s - loss: 0.8986 - accuracy: 0.8969
Epoch 8: val_accuracy improved from 0.94727 to 0.95703, saving model to ./bestmodel.h5
```

- Check Point File



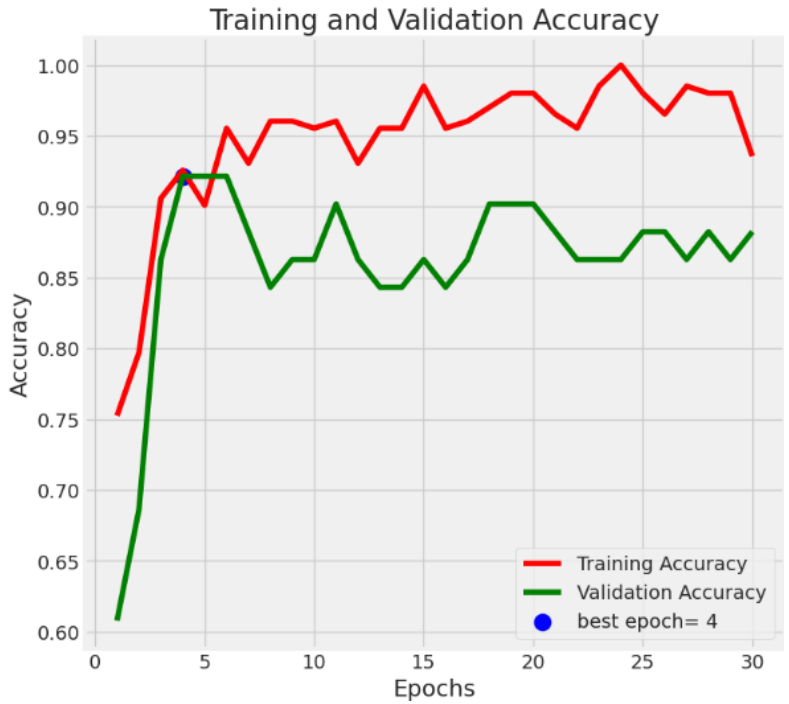
III. Model Evaluation

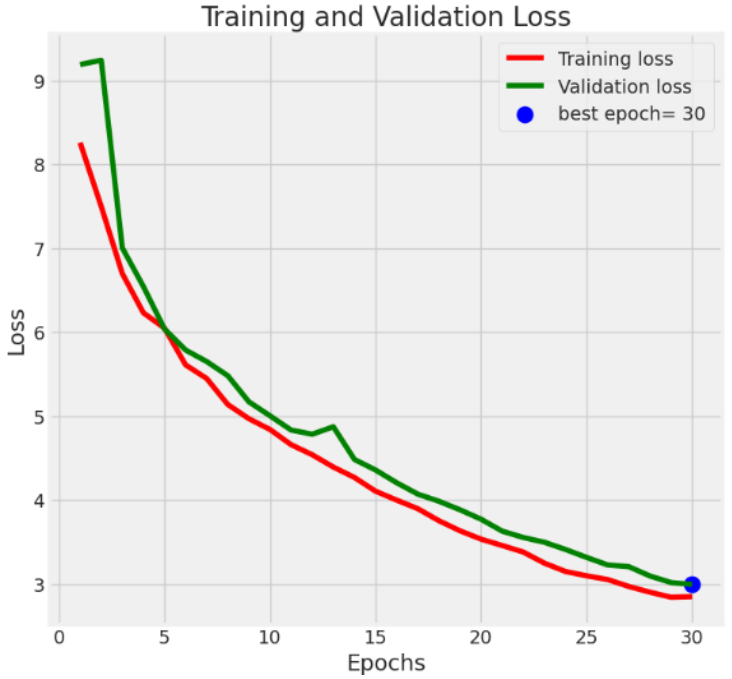
- Check Train and validation accuracy of the model.

```
train_score = model.evaluate(train_gen, verbose= 1)
test_score = model.evaluate(test_gen, verbose= 1)
```

```
--------------------
Test Loss:  2.99277925491333
Test Accuracy:  0.8823529481887817
```

- Train and Test (accuracy &loss)

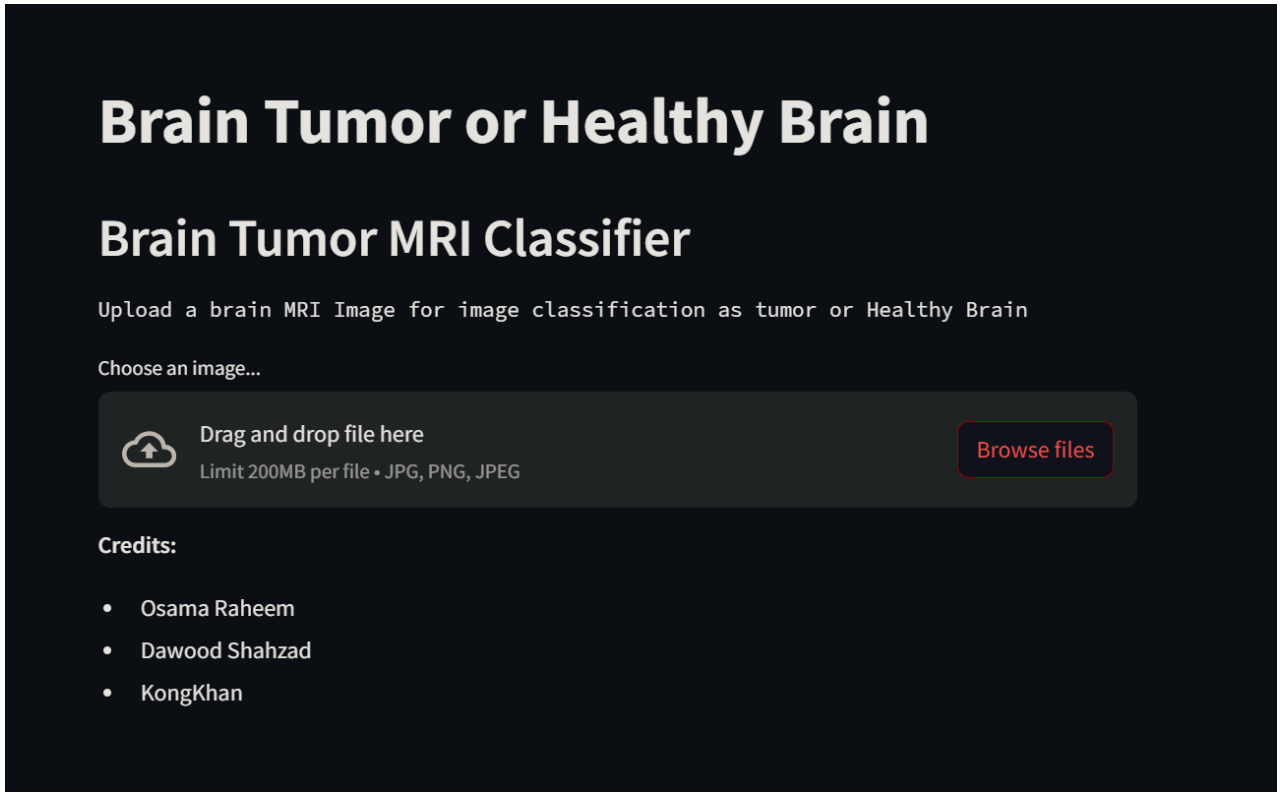




IV. Web App (Live Demo)

- We developed a web app for this project using Streamlit.
- Streamlit is a free and open-source framework to rapidly build and share beautiful machine learning and data science web apps.
- It is a Python-based library specifically designed for machine learning engineers.
- Screenshots :

Home Page





Conclusion

- The CNN-based approach for brain tumor detection showed high accuracy and robustness in detecting tumors from MRI images.

- Impact of the proposed solution:
- The solution can significantly aid radiologists in the early detection of brain tumors, improving diagnostic efficiency and patient care.

- Final thoughts:
- The project demonstrates the potential of deep learning in medical image analysis and its practical applications in healthcare.

Limitations

- Discussion of the limitations of the current solution:
- 1. Dependence on the quality and variety of training data.
- 2. The model's performance may vary with different MRI machines and imaging techniques.

- Potential challenges or constraints faced during the project:
- 1. Limited availability of labeled MRI datasets.
- 2. Balancing model complexity with computational resources.



Future Directions

Suggestions for future improvements:

1. Incorporate more diverse datasets to improve model generalization.

2. Explore the use of advanced CNN architectures and transfer learning.

Long-term vision related to the project:

To develop a comprehensive diagnostic tool that can detect various types of brain anomalies.

Potential areas for further research:

1. Integration with hospital information systems for real-time analysis and feedback.

2. Development of explainable AI techniques to enhance model interpretability for medical professionals.

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